

K210-Color Reaction

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1. Opening Notes

Please read the "Introduction to Motors and Usage" in the information of the four-way motor driver board first to understand the motor parameters, wiring methods, and power supply voltage you are currently using. To avoid burning the motherboard or motor.

Motor: The case and code take the 310 motor of our store as an example.

2. Experimental Preparation

National Race Chassis V2 Four-wheel Drive Version, 4*310 Motors, 7.4V Lithium Battery, K210 Vision Module, Light Bar, STM32F103C8T6 Core Board.

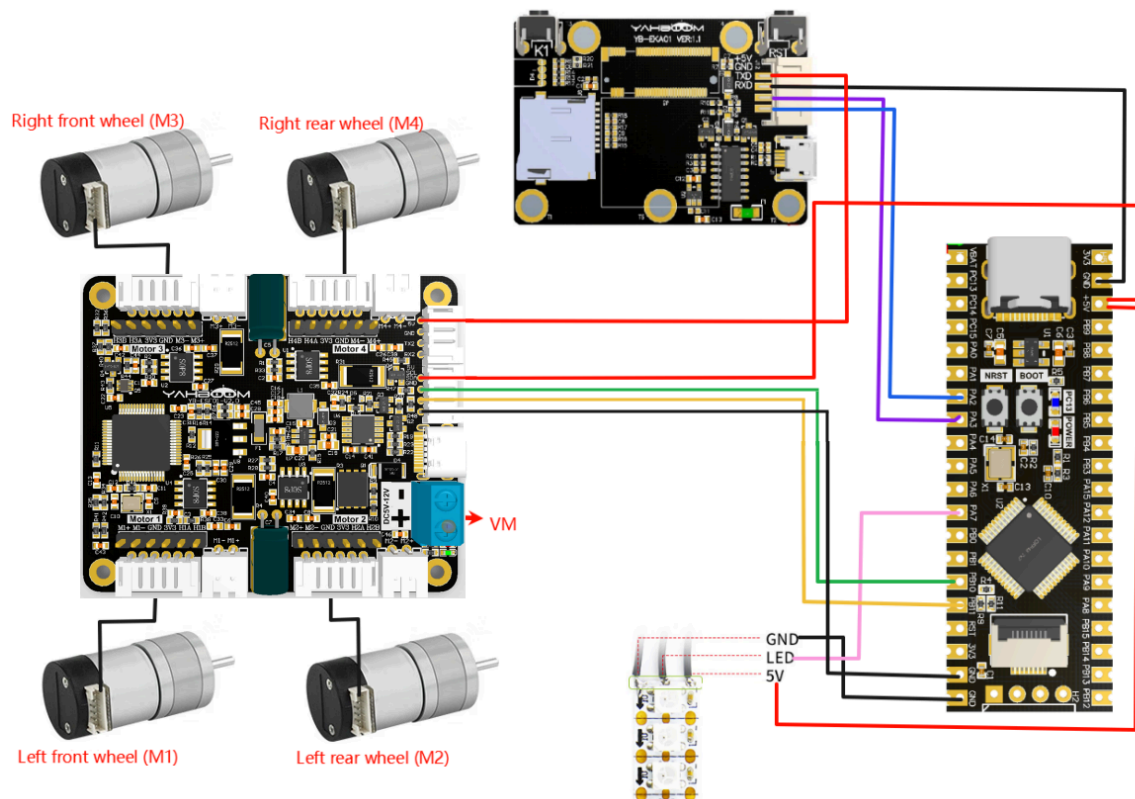
The relationship between the 4 motor interfaces and the car is as follows:

- M1 -> upper left motor (left front wheel of the car)
- M2 -> lower left motor (left rear wheel of the car)
- M3 -> upper right motor (right front wheel of the car)
- M4 -> lower right motor (right rear wheel of the car)

Hardware wiring:

Overall wiring

This case does not have a drive motor, so the motor can be left unconnected.



Wiring pins

Four-way motor driver board	STM32C8T6
5V	5V
GND	GND
SCL	PB10
SDA	PB11

Below, take the M1 motor as an example, and other motors are similar

Motor	Four-way motor driver board (Motor)
M2	M1-
VCC	3V3
A	H1A
B	H1B
GND	GND
M1	M1+

K210	STM32C8T6
5V	5V
GND	GND
TXD	PA3
RXD	PA2

3. Key code analysis

- bsp.c

```
#include "bsp.h"
void BSP_init(void)
{
    SystemInit();
    NVIC_PriorityGroupConfig(NVIC_PriorityGroup_2);

    delay_init();
    // delay_ms(1000); //等待红外稳定    wait for infrared to stabilize

    USART1_init(115200);
    USART2_init(115200); //使用串口2 接收红外    Use serial port 2 to receive infrared
    RGB_Init();
    IIC_Motor_Init(); //四路电机通信初始化    Four-way motor communication initialization

    //放到最后才生效，不然还是无法正常使用    It will take effect at the end, otherwise it will
    not work properly.
    RCC_APB2PeriphClockCmd(RCC_APB2Periph_AFIO, ENABLE);
    GPIO_PinRemapConfig(GPIO_Remap_SWJ_JTAGDisable, ENABLE); //禁用jlink 只用SWD调试口，PA15、
    PB3、4做普通IO    Disable jlink and use only SWD debug port, PA15, PB3, 4 as normal IO

}
```

The `BSP_init()` function is the key entry point for system initialization. First, the system clock is initialized through `SystemInit()`, and then the interrupt priority group is configured as group 2. Then the delay function, serial port (USART1 is used for debugging, USART2 is used to receive infrared signals), RGB light module, and IIC communication interface for four-way motor control are initialized in turn. Finally, the multiplexing function is enabled and JTAG is remapped to SWD, and the PA15, PB3, and PB4 pins are released for use as ordinary GPIOs to ensure that all peripherals work normally and are ready to run the main application.

- revaction.c

```
// 获取命令标志    Get command flag
uint8_t Get_CMD_Flag(void)
{
    return New_CMD_flag;
}
```

```

...

// 清除命令数据和相关标志 clear command data and related flags
void Clear_CMD_Flag(void)
{
    #if ENABLE_CLEAR_RXBUF
    for (uint8_t i = 0; i < New_CMD_length; i++)
    {
        RxBuffer[i] = 0;
    }
    #endif
    New_CMD_length = 0;
    New_CMD_flag = 0;
}

...

//数据分析 data analysis
void Upper_Data_Parse(uint8_t *data_buf, uint8_t num)
{
    uint8_t func_id = *(data_buf + 3);
    switch (func_id)
    {
        /* 判断功能字：小车速度设置 Judgment function word: Trolley speed setting */
        case FUNC_MOTION:
        {
            ...
        }

        /* 判断功能字：彩灯控制 Judgment function word: color light control */
        case FUNC_RGB:
        {
            u8 index = *(data_buf + 4);
            u8 red = *(data_buf + 5);
            u8 green = *(data_buf + 6);
            u8 blue = *(data_buf + 7);
            // printf("RED:%d, G:%d, B:%d\r\n",red,green,blue);
            RGB_Set_Color(index, red, green, blue);
            RGB_Update();
            break;
        }

        default:
            break;
    }
}

```

The `Get_CMD_Flag` function returns the current command flag `New_CMD_flag`, which is used to check whether there is a new command; the `Clear_CMD_Flag` function clears the command flag and the data in the receiving buffer.

The `Upper_Data_Parse` function parses the data sent by the upper layer, performs different operations according to different function words (`func_id`), and `FUNC_MOTION` extracts the light index and the values of the three RGB color channels in the command, and calls `RGB_Set_Color()` to set the color, and then uses `RGB_Update()` to update the display to realize the color light control function.

- main.c

```
#define MOTOR_TYPE 2    //1:520电机 2:310电机 3:测速码盘TT电机 4:TT直流减速电机 5:L型520电机
                        //1:520 motor 2:310 motor 3:speed code disc TT motor 4:TT DC
                        reduction motor 5:L type 520 motor

int main(void)
{

    //硬件初始化 Hardware Initialization
    BSP_init();

    #if MOTOR_TYPE == 1
        ...

    #elif MOTOR_TYPE == 2
        Set_motor_type(2); //配置电机类型 Configure motor type
        delay_ms(100);
        Set_Pluse_Phase(20); //配置减速比 查电机手册得出 Configure the reduction ratio. Check the
motor manual to find out
        delay_ms(100);
        Set_Pluse_line(13); //配置磁环线 查电机手册得出 Configure the magnetic ring wire. Check the
motor manual to get the result.
        delay_ms(100);
        Set_wheel_dis(48.00); //配置轮子直径,测量得出 Configure the wheel diameter and measure
it
        delay_ms(100);
        Set_motor_deadzone(1300); //配置电机死区,实验得出 Configure the motor dead zone, and the
experiment shows
        delay_ms(100);

        ...
    #endif

    while(1)
    {
        if (Get_CMD_Flag())
        {
            Upper_Data_Parse(Get_RxBuffer(), Get_CMD_Length());
            Clear_CMD_Flag();
        }
    }
}
```

MOTOR_TYPE: used to set the motor type to be used. Modify the corresponding number according to the comments based on the motor you are currently using.

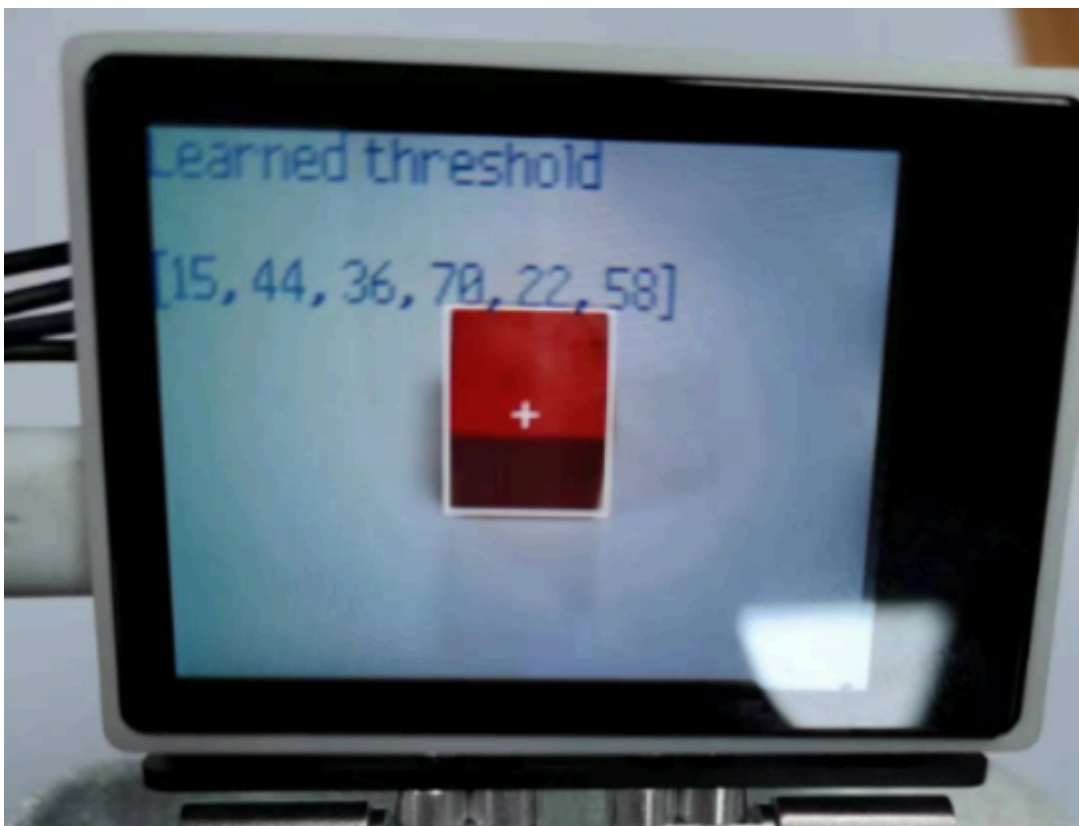
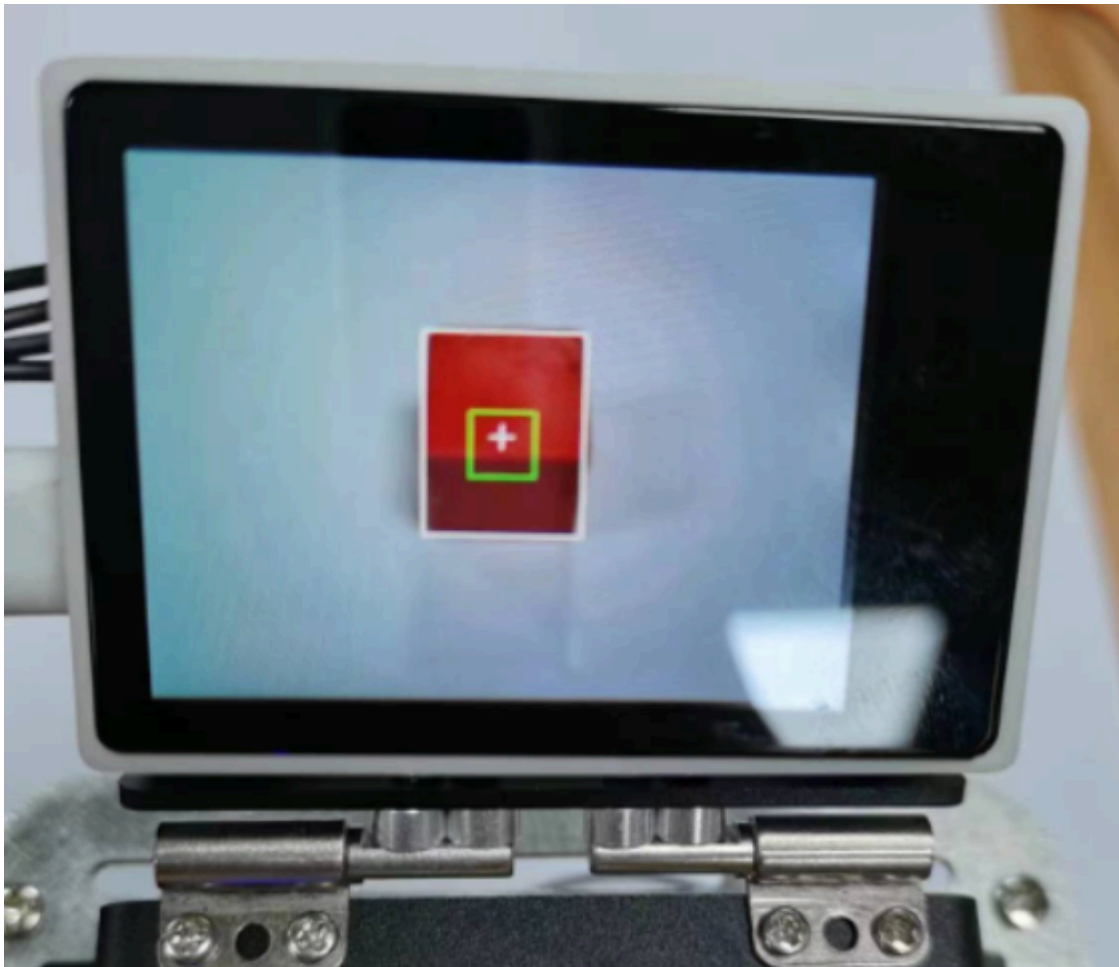
Call the `BSP_init()` function to initialize the hardware settings, and use `Set_Motor(MOTOR_TYPE)` to set the motor type and parameters. In the `while(1)` loop, `Get_CMD_Flag` detects whether there is a new command. If so, call `Upper_Data_Parse` to parse the data and set the RGB light color. After processing, use `Clear_CMD_Flag` to `clear the command flag` to ensure that the system continues to respond to the host computer instructions.

4. Experimental operation

1. Burn the program to STM32.
2. Download the car drive library and PID control library in the K210\library directory to the root directory of the memory card in advance.
3. Open the `color_rgb.py` code with CanMV IDE.
4. Please uncomment `learning_color`, then run the program again in CanMV IDE, put the color to be recognized into the box for learning, and wait for the learning to be completed. The serial terminal and display at the bottom of the IDE will display the recognized color value, and replace the recognized color value with the corresponding color value.

```
learning_color()

#红色 color red
threshold_red = [30, 59, 42, 69, 19, 47]
#绿色 color green
threshold_green = [30, 72, -39, 2, 20, 44]
#蓝色 color blue
threshold_blue = [34, 73, 2, 54, -75, -40]
#黄色 color yellow
threshold_yellow = [71, 99, -3, 27, 54, 67]
```



5. Run the program four times in sequence, and update the values of `threshold_red`, `threshold_green`, `threshold_blue`, and `threshold_yellow` in sequence. After color learning is completed, comment out the `learning_color` function again.

```
#learning_color()

#红色 color red
threshold_red = [30, 59, 42, 69, 19, 47]
#绿色 color green
threshold_green = [30, 72, -39, 2, 20, 44]
#蓝色 color blue
threshold_blue = [34, 73, 2, 54, -75, -40]
#黄色 color yellow
threshold_yellow = [71, 99, -3, 27, 54, 67]

color_list = [threshold_red, threshold_green, threshold_blue, threshold_yellow]
```

6. Download the program into the K210 module and run it.
7. Connect all the wires of the car.
8. Place the car in a white background, move the K210 module bracket to a suitable angle, and connect the power supply.

5. Experimental phenomenon

After waiting for the system to initialize, please put a red, green, blue or yellow color block in front of the camera's view. The car will automatically light up the corresponding color according to the color of the current color block. If there are multiple color blocks, the car's RGB light will show a marquee effect.